

ABHISHEK RAJESH KHARAT

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[GitHub](#) • [LinkedIn](#) • [Medium](#) • [Hugging Face](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Pre-final year engineering student specializing in AI/ML, Cloud Computing, and Agentic Systems. Experienced in building and deploying multi-agent Reinforcement Learning environments, production-grade FastAPI backends, and LangGraph workflow agents with Human-in-the-Loop escalation. Proven track record of developing automated FinOps and DevOps solutions. Actively seeking AI/ML or data engineering internships.

PROFESSIONAL EXPERIENCE

Agentic AI Intern | Innomatics Research Labs | Remote

Feb 2026 – May 2026

- Developed and deployed LangChain and LangGraph agent pipelines for multi-step AI task automation, improving task execution flow.
- Built modular backend services using FastAPI and integrated them with AWS cloud infrastructure (Amazon Bedrock).
- Published a narrative-driven technical blog on Medium explaining NLP preprocessing pipelines with custom SVG visuals.
- Completed structured training in Python programming, Generative AI agent design patterns, and prompt engineering.

Artificial Intelligence Intern | Codtech IT Solutions | Remote

Jan 2026 – Feb 2026

- Developed an NLP-based Text Summarization Tool comparing TF-IDF baselines and fine-tuned BART transformer models via Hugging Face.
- Built a Speech-to-Text Recognition System utilizing the SpeechRecognition library, Google Speech API, and OpenAI Whisper.
- Implemented a Neural Style Transfer pipeline using a pre-trained VGG19 CNN in PyTorch to apply artistic styles to target images.
- Created a Generative Text Model utilizing GPT-2 for automated, topic-based paragraph generation.

PROJECTS

AI Tribunal – Adversarial Legal RL Environment

Meta Hackathon Finalist

[GitHub Repository](#) • [Hugging Face Space](#)

- Designed and implemented a custom Gymnasium reinforcement learning environment to model multi-agent strategic interactions, hidden information, and deception detection.
- Trained deep RL policies (PPO/DQN) using PyTorch, optimizing policy neural networks to identify and fact-check fabricated evidence in sparse-reward environments.
- Developed a web-based evaluation interface deployed on Hugging Face Spaces to benchmark AI agents and human performance under deceptive legal scenarios.

OrbitWatch – Agentic Space Debris Intelligence System

FAR AWAY 2026

[GitHub Repository](#) • [Live Dashboard](#)

- Developed a space situational awareness platform utilizing SGP4 orbital propagation algorithms to calculate real-time conjunction risk from raw Two-Line Element (TLE) satellite data.
- Integrated the OpenAI Responses API to build an agentic briefing engine that translates complex orbital vectors and collision parameters into structured natural-language operator briefs.
- Built a responsive FastAPI/HTML dashboard with automated email alerts via Resend API and session state persistence using Upstash Redis.

ShipSage – AI-Assisted DevOps Readiness Tool

IBM Bob Hackathon

[GitHub Repository](#) • [Live Dashboard](#)

- Developed a DevOps copilot that analyzes GitHub repository structures to evaluate production readiness, score DevOps posture, and identify deployment blockers.
- Integrated the IBM watsonx Granite LLM with custom deterministic fallback rules to generate multi-stage Dockerfiles, Kubernetes manifests, and AWS Terraform code tailored to the codebase.

- Designed a scalable FastAPI backend deployed on Render to parse repositories and dynamically generate Prometheus/Grafana monitoring configurations and CI/CD pipelines.

EDUCATION

Bachelor of Engineering (B.E.) in Automation & Robotics | MES Wadia College of Engineering, Pune **Expected May 2027**

TECHNICAL SKILLS

Languages: Python, C++ (basic), Bash (basic)

AI / ML: LangChain, LangGraph, Gymnasium (RL), NLTK, spaCy, PyTorch, ChromaDB, Embeddings, Prompt Engineering

Cloud & DevOps: AWS (Bedrock, QuickSight), Terraform (IaC), Docker, Kubernetes, CI/CD, Git, GitHub, HF Spaces, Vercel

Web & APIs: FastAPI, Flask, REST APIs, React (basic)

Tools: Linux/Ubuntu, Jupyter, VS Code

ACHIEVEMENTS

- Meta/PyTorch OpenEnv Hackathon 2026 — Finished as finalist (top 800 of 31,000+ teams); designed and trained AI Tribunal RL environments.
- FAR AWAY 2026 Hackathon (Space & Aerospace theme) — Developed OrbitWatch solo (team HallucinateThis); built agentic space debris conjunction tracking.
- lablab.ai Web Data UNLOCKED Hackathon — Developed Cloud Waste Sniper (team HallucinateThis) for automated Terraform FinOps.
- IBM Bob Hackathon — Developed ShipSage/RepoSage with a teammate; built AI-powered DevOps and Kubernetes manifest generator.
- Technical Author on Medium — Published articles explaining NLP preprocessing pipelines with custom SVG visuals.